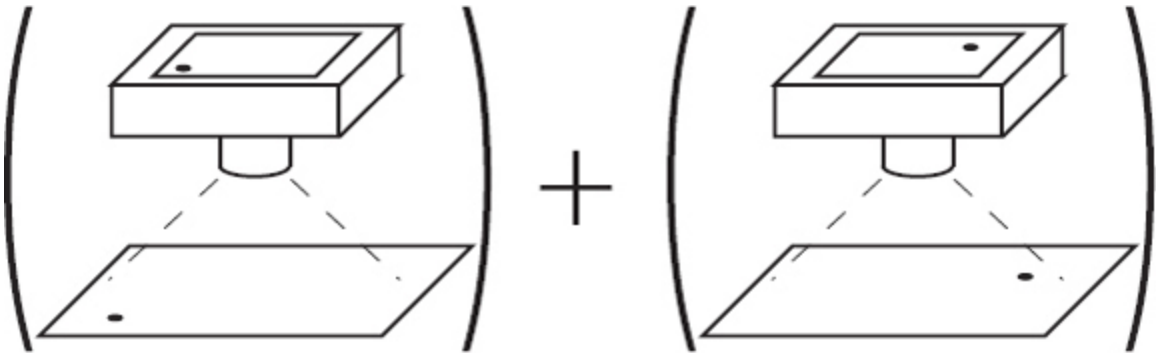
interaction, we might expect that the camera itself is now in a superposition of all the possible measurement outcomes it might have observed: the camera saw the electron in *this* location, or the camera saw the electron in *that* location, and so on.

If that were the whole story, AQM would be an untenable mess. Electrons in superpositions, cameras in superpositions, nothing much resembling the robust approximately classical world of our experience.

Fortunately we can appeal to another startling feature of quantum mechanics: given two different objects (like an electron and a camera), they are not described by separate, individual wave functions. There is *only one wave function*, which describes the entire system we care about, all the way up to the "wave function of the universe" if we're talking about the whole shebang. In the case under consideration, there is a wave function describing the combined electron+camera system. So what we really have is a superposition of all possible combinations of where the electron might have been located, and where the camera actually observed it to be.

Although such a superposition in principle includes every possibility, most of the possible outcomes are assigned zero weight in the quantum state. The cloud of probability vanishes into nothingness for most possible combinations of electron location and camera image. In particular, there is no probability that the electron was in one location but the camera saw it somewhere else (as long as you have a relatively functional camera).



This is the quantum phenomenon known as *entanglement*. There is a single wave function for the combined electron+camera system, consisting of a superposition of various possibilities of the form "the electron was at this location, and the camera observed it at the same location." Rather than the electron and the camera doing their own thing, there is a connection between the two systems.